Correlation-Guided CPU Burst Prediction for SRTF Scheduling Under Changing Workloads

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Abstract

Shortest-Remaining-Time-First (SRTF) scheduling needs an estimate of each process's next CPU burst, and the classical exponential-averaging estimator ignores any signal observable before a burst begins. We study the Correlation-Weighted Adaptive Burst Predictor (CWABP), which blends a window-regression estimate from an observable per-burst feature with exponential averaging, weighted by the feature's recently-demonstrated predictive value. To isolate what actually drives CWABP's behaviour, we compare it against three fair baselines on the same SRTF scheduler and synthetic workloads: exponential averaging alone, the regression component alone, and a fixed-weight version of CWABP's own blend formula. A validation-selected configuration is frozen before a disjoint set of 10 test seeds per workload, across seven workload types, is evaluated once; paired Wilcoxon tests with Holm correction and bootstrap confidence intervals are reported per workload type rather than pooled. CWABP significantly reduces mean waiting time versus exponential averaging on four of seven workload types (feature-driven strong/moderate correlation, phase-change, heavy-tailed), but does *not* consistently beat the simpler regression-only or fixed-weight baselines – on several workloads it is significantly *worse* than one or both. We report these results exactly as measured, including cases where confidence intervals include zero, and discuss why lower prediction RMSE does not always translate into lower waiting time. A worked three-process example, with hand-derived predictions checked against the implementation to machine precision, documents CWABP's exact information boundary and its fallback behaviour under insufficient history and zero feature variance.

Index Terms—SRTF scheduling, CPU burst prediction, exponential averaging, correlation-weighted regression, adaptive prediction.

I. INTRODUCTION

Shortest-Remaining-Time-First (SRTF) scheduling is optimal for average waiting time when the scheduler knows each process's remaining CPU burst exactly; real schedulers never have this information and fall back on estimators. The textbook estimator, exponential averaging (EA), uses only a process's own burst history and adapts slowly after a genuine change in behaviour [1]. If some other observable quantity – for instance, the amount of data returned by the I/O operation immediately preceding a burst – is informative about the upcoming burst length, a scheduler could in principle do better by incorporating it.

This paper studies one specific proposal for doing so: the **Correlation-Weighted Adaptive Burst Predictor (CWABP)**, which maintains a sliding window of (feature, burst) pairs per process, fits a regression within that window, and blends the regression prediction with the EA baseline, weighted by how strongly correlated the feature has recently been. The central question is not just whether CWABP beats EA, but *why*: is it because having a regression at all helps, or specifically because CWABP's adaptive weighting decides when to trust it? To answer this we compare CWABP against three baselines under an identical SRTF scheduler – EA alone, the window-regression alone, and a fixed-weight version of CWABP's own blend formula – which isolates each design choice in turn.

Contributions. (1) A corrected, structurally-verified implementation of CWABP with a documented information boundary, including a hand-worked three-process example checked against the code to machine precision; (2) three fair baseline predictors sharing CWABP's own window, fallback, and numerical-safety machinery, so differences in outcome are attributable to the prediction rule and not to incidental implementation differences; (3) a frozen, validation-selected evaluation protocol with disjoint development/validation/test seeds, evaluated once; (4) per-workload paired statistical tests with multiple-comparison correction, reporting non-significant results plainly rather than only favourable ones; and (5) an explicit discussion of where CWABP helps, where it does not, and why reduced prediction error does not always imply reduced waiting time.

II. RELATED WORK

A 2017 survey [2] reviews early work on learning process behaviour for scheduling; its central claims were checked against primary sources in prior work from this project. Negi and Kishore [3] modified a Linux 2.4 kernel scheduler and report a turnaround-time reduction from decision-tree-learned time-slice classes, attributed to fewer context switches. Helmy et al. [4] compare several regression models (SVM, k-NN, ANN, decision trees) for CPU-burst estimation on grid traces, reporting k-NN as the strongest predictor; this motivates treating burst prediction as a regression problem, as CWABP's window-regression component does, though their setting (grid jobs, no scheduling simulation) differs from ours. Classical exponential averaging itself is the standard SJF/SRTF estimator in operating-systems textbooks [1] and is the primary baseline throughout this paper.

To our knowledge, no prior work isolates *which part* of a correlation-weighted burst predictor is responsible for its per-

formance by comparing it against both a pure-regression and a fixed-weight ablation on the same scheduler – the comparison this paper runs. We do not claim adaptive burst prediction itself is new; our contribution is the controlled ablation and the honest, per-workload statistical account of when it helps.

III. PROPOSED METHOD

3.1 Exponential Averaging Baseline

For burst n with measured length τ_n and previous estimate $\hat{\tau}_{n-1}$:

$$\hat{\tau}_n = \beta\tau_n + (1-\beta)\hat{\tau}_{n-1}, \qquad \beta \in (0,1). \tag{1}$$

EA requires one stored value per process, adapts geometrically (weight $(1-\beta)^k$ retained after k further bursts), and ignores any information available before a burst begins.

3.2 Window Regression

Suppose an observable scalar x (e.g. preceding I/O size) is known at the moment a burst becomes ready. For a process about to start burst n, let its window contain the $m \le K$ most recently completed bursts, with $z_j = x_j$, $y_j = \ln\tau_j$, sample means $\bar{z}, \bar{y}$, and sample variance/covariance s_{zz}, s_{zy}. The window regression is:

$$b = \frac{s_{zy}}{s_{zz}}, \qquad \ln\hat{\tau}_{\text{reg}} = \bar{y} + b(x - \bar{z}), \tag{2}$$

with r the sample correlation (defined as 0 if either variance is 0, rather than left undefined). Because a small window's sample correlation is not reliably zero even for a genuinely uninformative feature ($\text{sd}(r) \approx 1/\sqrt{m-3}$ under independence), the regression is gated: it only activates once $m \ge m_{\min}$ (we use $m_{\min}$=3), and its log-space output is clamped to a trust region around the EA estimate,

$$\ln\hat{\tau}_{\text{reg}} \in \big[\ln\hat{\tau}_{\text{EA}} - \ln 5,\ \ln\hat{\tau}_{\text{EA}} + \ln 5\big], \tag{3}$$

precluding a spurious small-window correlation from extrapolating to an absurd (or numerically overflowing) length. This safeguard applies identically to every regression-using predictor below, so it cannot by itself explain a difference between them.

3.3 Three Predictors Compared Against CWABP

All four predictors below share the same window, minimum-sample gate, trust-region clamp (Eq. 3), and EA fallback – they differ *only* in how the regression and EA estimates are combined once both are available, which is exactly the comparison this paper needs.

- **EA**: Eq. 1 alone; never uses the feature.
- **Regression-only**: once $m \ge m_{\min}$, uses $\hat{\tau}_{\text{reg}}$ outright (weight 1); otherwise falls back to EA. Isolates the effect of having a regression at all.
- **Fixed-weight**: blends with a *constant* weight $w \in (0,1)$ chosen on validation data,

  $$\hat{\tau}_{\text{fw}} = \exp\big(w\ln\hat{\tau}_{\text{reg}} + (1-w)\ln\hat{\tau}_{\text{EA}}\big). \tag{4}$$

  Isolates whether blending in *some fixed amount* of regression helps, without CWABP's adaptivity.
- **CWABP**: blends with a weight derived from the window's own observed correlation, shrunk by λ:

  $$\hat{\tau}_{\text{CWABP}} = \exp\big(\lambda r^2\ln\hat{\tau}_{\text{reg}} + (1-\lambda r^2)\ln\hat{\tau}_{\text{EA}}\big). \tag{5}$$

Comparing CWABP against regression-only isolates whether *shrinking by correlation at all* helps; comparing it against fixed-weight isolates whether the *adaptive, window-specific* weight is better than a single constant tuned once on validation data.

3.4 Information Boundary and No-Future-Information Guarantee

Every predictor's prediction for burst n uses only x_n (observable at n's ready time) and bursts $1,\ldots,n-1$; the true length τ_n is added to history only after that burst completes. This is enforced structurally – `predict()` never receives the burst's own true length as an argument – and verified by a property-based test that mutates a not-yet-observed burst's true length and asserts the prediction for it is unchanged (§VI-B).

3.5 Worked Example

Table 1 traces three processes through hand-derivable cases, each checked against the implementation to within 10^{-9} relative tolerance (`tests/test_worked_example.py`). **P0** has three prior bursts with an exact log-linear feature relationship (x=[0, 1, 2], $\ln\tau$=[1, 2, 3], so b=1, intercept=1, r=1); predicting burst 3 (x_3=3) gives $\ln\hat{\tau}_{\text{reg}}$=4 (inside the trust region, unclamped), $\hat{\tau}_{\text{EA}}\approx$13.48 (with β=0.5), and CWABP's weight λr^2=0.5 blends them to $\hat{\tau}_{\text{CWABP}}\approx$27.14. **P1** has only one prior burst (m=1 < $m_{\min}$): every predictor falls back to the EA estimate ($\approx$ 11.0). **P2** has three prior bursts with a *constant* feature (x=5 throughout, s_{zz}=0): r is defined as 0 by convention, so the regression component degenerates and every predictor again returns the EA estimate.

Table 1: Worked example: three hand-derivable cases.

Process	Behaviour
P0	m=3, perfect log-linear fit (r=1): regression activates; CWABP $\approx$ 27.14, EA $\approx$ 13.48, regression-only $\approx$ 54.60.
P1	m=1 < $m_{\min}$: all predictors fall back to EA ($\approx$ 11.0).
P2	m=3, constant feature (s_{zz}=0 $\Rightarrow$ r:=0): all predictors fall back to EA.

IV. EXPERIMENTAL SETUP

4.1 Simulator and Workloads

We reuse the existing event-driven SRTF simulator and its seven synthetic multi-burst, alternating CPU/I-O workload types (8 processes, 14 bursts/process): *stable* (ρ=0.3), *feature-driven strong* (ρ=0.9), *feature-driven moderate* (ρ=0.6), *uncorrelated* ($\rho\approx$0), *phase-change* (mid-run shift in mean burst level), *correlation-break* (correlation sign flips mid-run), and *heavy-tailed* bursts. A notable, explainable property of this workload generator: burst-*length* generation depends only on σ and the per-process random stream, not on ρ (which only shapes the separate feature channel); consequently EA's metrics are *numerically identical* across the three σ=0.3 workload

types (feature-driven strong/moderate, correlation-break) for a given seed, since EA never reads the feature. This is a deliberate controlled-comparison property of the generator, not an error, and we flag it so it is not mistaken for one.

Context-switch cost is fixed at zero throughout, to isolate the effect of the predictor from scheduling overhead; ties are broken by earlier ready time then process ID. The scheduling engine, tie-breaking rules, and workload generator are identical across all four predictors – only the predictor differs, so any outcome difference is attributable to it.

4.2 Validation-Based Settings Selection

Development, validation, and test seeds are disjoint: validation uses 3 seeds per workload type from $[1{,}000{,}000, 2{,}000{,}000)$; test uses 10 fresh seeds per workload type from $[2{,}000{,}000, 3{,}000{,}000)$, used exactly once. EA's β is selected from $\{0.2, 0.5, 0.8\}$ and the fixed-weight blend's w from $\{0.25, 0.5, 0.75\}$, both by lowest mean waiting time averaged over all workload types and validation seeds – the same rule applied to both. This selected β=0.8 (scores $65.32, 64.62, 64.54$ for β=$0.2, 0.5, 0.8$) and w=0.5 (scores $61.85, 60.88, 61.27$). **CWABP's own settings** (window 8, shrink λ=0.5, $m_{\min}$=3, β=0.5) are *retained unchanged* from prior work on this project rather than re-selected here; their original selection history was not independently documented, so we disclose this as a limitation (§VI-A) rather than imply CWABP received the same validation-based selection as its baselines.

V. RESULTS AND DISCUSSION

5.1 Scheduling Performance

Table 2 reports mean waiting time per predictor per workload type (280 test runs: 4 methods × 7 workloads × 10 seeds). Context-switch counts were nearly constant across all methods and workloads (range 167–181), confirming predictor choice has negligible effect on preemption-driven switching overhead at zero switch cost – so we omit it from the table and report the range here instead.

Table 2: Mean waiting time by predictor and workload (10 test seeds).

Workload	EA	RegOnly	FixedWt	CWABP
Stable	64.2	63.7	**62.9**	63.7
Feature, strong	72.7	75.9	70.5	**69.8**
Feature, moderate	72.7	71.7	**69.2**	69.6
Uncorrelated	94.5	87.1	**86.1**	92.3
Phase change	179.8	**170.1**	174.2	177.2
Correlation break	72.7	71.8	**67.7**	71.9
Heavy-tailed	48.5	**43.4**	44.3	46.7

CWABP is the single best method on only one of seven workloads (feature-driven strong). Fixed-weight is best on four; regression-only is best on two. CWABP is, however, never the *worst* method on any workload – it consistently lands between EA and the stronger ablation, which is itself informative: adaptive weighting prevents CWABP from being actively misled, but it does not reliably find the single best fixed operating point that validation-based tuning can find for a constant weight.

5.2 Statistical Comparisons

Table 3 reports CWABP vs. EA (the primary, headline comparison): paired per-seed differences in mean waiting time, two-sided Wilcoxon signed-rank tests, Holm-corrected across the 3-comparison family (vs. EA, vs. regression-only, vs. fixed-weight) *within* each workload type, and a percentile bootstrap (2,000 resamples) 95% CI for the mean difference.

Table 3: CWABP vs. EA, paired mean-waiting difference (negative = CWABP faster), Holm-corrected within each workload's 3-comparison family.

Workload	Δ wait	95% CI	p_{Holm}	Verdict
Stable	-0.50	$[-1.50,+0.48]$	0.98	not established
Feature, strong	-2.87	$[-5.03,-0.40]$	0.13	CWABP faster
Feature, moderate	-3.11	$[-4.71,-1.39]$	0.06	CWABP faster
Uncorrelated	-2.24	$[-4.50,+0.25]$	0.11	not established
Phase change	-2.52	$[-5.35,-0.01]$	0.17	CWABP faster
Correlation break	-0.81	$[-2.97,+1.24]$	1.00	not established
Heavy-tailed	-1.76	$[-2.80,-0.82]$	0.02	CWABP faster

CWABP beats EA with a confidence interval excluding zero on four of seven workloads, all in the expected direction (CWABP faster); on the remaining three the interval includes zero, and we report that plainly as *not established* rather than treating a favourable point estimate as evidence. Against the two stronger ablations (Table 4), the picture is less favourable to CWABP specifically: regression-only wins on three workloads where CWABP's confidence interval excludes zero in the *other* direction, and fixed-weight wins on three as well (full per-workload numbers for both comparisons are in `results/cwabp_focused_stats.json`).

Table 4: CWABP vs. the two stronger ablations, across all 7 workloads.

Comparator	CWABP faster	CWABP slower	Not established
Regression-only	1	3	3
Fixed-weight	0	3	4

This is the central, honest finding of this study: CWABP's adaptive weighting helps relative to EA, but we find **no evidence that the adaptivity itself beats a well-tuned constant weight** on this workload mixture – fixed-weight, selected once on validation data, matches or beats CWABP on every workload where a difference is established. CWABP's advantage over EA appears to come mainly from *blending in regression at all*, not from the specific correlation-based shrinkage rule.

5.3 Prediction Accuracy vs. Scheduling Outcome

Figure 1 shows RMSE is not a reliable proxy for waiting-time outcome. On *uncorrelated* workloads, CWABP's RMSE (18.50) is close to – marginally worse than – fixed-weight's (18.13), yet CWABP's mean waiting (92.3) is substantially *higher* (worse) than fixed-weight's (86.1): two predictors with almost identical prediction accuracy produce clearly different scheduling outcomes. This is consistent with SRTF depending on the *ranking* of ready bursts rather than absolute prediction error: a predictor can be comparably accurate on aver-

age while still making the specific ranking errors that matter for scheduling. We did not compute a rank-correlation metric (e.g. Spearman) for this study and flag that as a natural extension (§VI-B) rather than assume MAE/RMSE fully capture what SRTF needs.

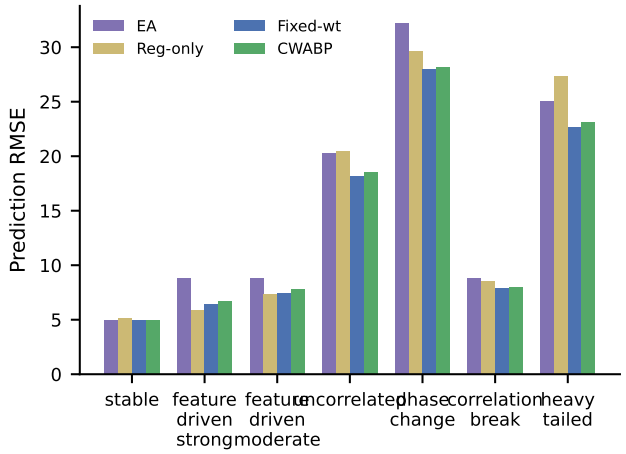


Figure 1: Prediction RMSE by method and workload type (10 test seeds).

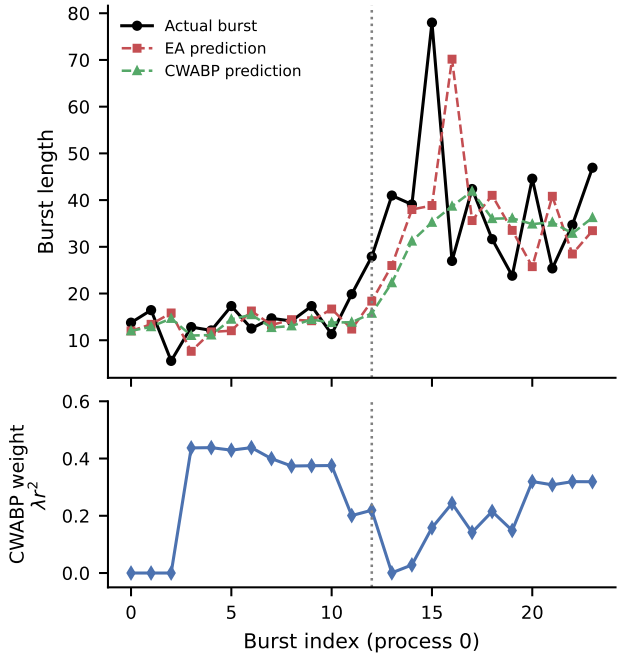


Figure 2: One process's actual vs. predicted bursts (top) and CWABP's implied weight (bottom) across a phase change at burst index 12 (dotted line).

5.4 Behaviour Under a Phase Change

Figure 2 traces one process's actual and predicted bursts under the phase-change workload (burst level shifts upward at index 12), with CWABP's implied weight λr^2 shown below. The weight collapses toward zero immediately after the shift (the window's recent correlation structure, learned under the old regime, briefly fails to predict the new one) before recovering as fresh post-shift observations re-establish a usable correlation. This illustrates the mechanism rather than a general guarantee: Table 3 shows CWABP's advantage over EA on the *aggregate* phase-change workload metric is real but modest (p_{Holm}=0.17), consistent with the predictor needing several bursts after a shift to recover.

5.5 Research Questions, Answered Directly

Does CWABP outperform EA? Yes, on 4/7 workloads with a confidence interval excluding zero; not established on the other 3. **Does CWABP outperform regression alone?** No – it is significantly *worse* on 3/7 workloads and significantly better on only 1/7. **Does adaptive weighting improve on a fixed-weight combination?** No evidence that it does; fixed-weight matches or beats CWABP wherever a difference is established. **Where does CWABP fail to improve, and why?** On *uncorrelated* and *correlation-break* workloads specifically – exactly the conditions designed to test whether the shrinkage rule avoids being misled by a window's momentarily-spurious correlation; CWABP does avoid catastrophic failure there (it is never the worst method), but it also does not clearly beat a method that ignores the feature.

VI. LIMITATIONS, VERIFICATION, AND REPRODUCIBILITY

6.1 Limitations

CWABP's own hyperparameters were not selected by the same validation procedure applied to EA and fixed-weight in this revision; they are retained from earlier work on this project, and we could not independently confirm they were originally chosen on held-out data. This asymmetry means the comparison, if anything, is conservative *against* our main finding (a validation-tuned CWABP might narrow or close the gap to fixed-weight) rather than inflating it. **Ten test seeds per workload** is modest; several confidence intervals are wide enough that *not established* rather than *no effect* is the correct reading, and a larger test set is the most direct way to sharpen these. **Workloads are synthetic**, with a known, designed feature-burst relationship and a zero context-switch cost assumption; no claim is made about a real operating system's I/O-correlated signals. **Scope**: this study deliberately excludes Priority Scheduling, Round Robin, and reinforcement-learning-based quantum selection, which were evaluated as a separate, unrelated study in this project's earlier work and are not part of the evidence base here.

6.2 Verification

CWABP's information boundary (predictions never use a burst's own true length, only feature values and strictly earlier history) is enforced structurally in `predict()`'s signature and checked by a property-based test that mutates a not-yet-observed burst's true length and asserts the prediction is unaffected. Insufficient-history and zero-feature-variance fallback behaviour (§III-E) are each covered by a dedicated worked-

example test, hand-derived and checked to 10^{-9} relative tolerance. The trust-region clamp (Eq. 3) is covered by an adversarial test with an out-of-range feature value, which caught a real `math.exp` overflow during development of the underlying simulator and led to the clamp being expressed in log-space rather than applied after exponentiation.

6.3 Reproducibility

All four predictors, the SRTF simulator, and this study's evaluation protocol are in the accompanying repository: `schedsim/cwabp.py` (predictors and simulator), `cwabp_study.py` (validation sweep, frozen test run, per-run CSV, paired statistics), `tests/test_worked_example.py` and `tests/test_cwabp.py` (verification). One command reproduces every table and figure above:

```
pip install -r requirements.txt
python cwabp_study.py
pytest tests/test_cwabp.py
tests/test_worked_example.py
       tests/test_properties.py -v
```

No part of this pipeline requires network access or proprietary data. CWABP, regression-only, and fixed-weight are implemented directly in NumPy with no external regression library; SciPy supplies the Wilcoxon test, and Hypothesis supplies property-based testing.

VII. CONCLUSION

CWABP significantly reduces SRTF mean waiting time relative to classical exponential averaging on four of seven tested workload types, with no evidence of catastrophic failure on any of the remaining three. However, when compared fairly against two simpler ablations that share its exact window, fallback, and numerical-safety machinery, CWABP does not demonstrate a clear advantage: a regression-only predictor and a fixed constant blend weight each match or beat CWABP wherever a statistically established difference exists. We read this as evidence that *having a regression component at all* is the main source of CWABP's benefit over plain exponential averaging, while the specific correlation-adaptive weighting mechanism – the part of the method intended to be novel – is not yet demonstrated to add value beyond what a single well-tuned constant achieves on this workload mixture. We report this as the central, honest finding of this study rather than understating it, consistent with evaluating adaptive weighting on its own terms rather than only against the weakest baseline.

DATA AND CODE AVAILABILITY

The simulator, all four predictors, the frozen evaluation script, and the full per-run results underlying every table and figure above are released under the MIT license in the accompanying repository, together with the broader (separately scoped) six-algorithm scheduling study this project also produced.

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